

KRANGEDE, Sweden

WMO: 022470

Lat: 63.1512N

Lon: 16.1699E

Elev: 184

StdP: 99.13

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.5	-23.5	-29.6	0.2	-26.4	-26.4	0.3	-23.5	7.1	-0.2	6.0	-0.3	0.3	330	0.629

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	25.8	16.9	23.9	16.2	21.9	15.3	18.3	23.5	17.2	21.9	16.1	20.4	2.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	12.0	20.1	15.2	11.1	18.8	14.2	10.4	17.9	52.6	23.4	48.9	22.0	45.8	20.4	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.4	(b) 5.4	(c) 4.7	(e) -29.1	(f) 28.4	(g) 2.9	(h) 1.9	(i) -31.2	(j) 29.7	(k) -32.9	(l) 30.9	(m) -34.6	(n) 31.9	(o) -36.7	(p) 33.3		
(a) 6.4	(b) 5.4	(c) 4.7	(e) -29.3	(f) 20.1	(g) 2.9	(h) 1.4	(i) -31.4	(j) 21.1	(k) -33.1	(l) 21.9	(m) -34.7	(n) 22.7	(o) -36.8	(p) 23.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.2	-8.0	-7.0	-3.0	2.6	8.1	12.3	15.2	13.5	9.1	3.1	-2.0	-6.4		
	DBStd	9.39	7.23	7.04	5.39	3.46	3.84	3.18	2.90	2.97	3.19	3.61	4.89	7.00		
	HDD10.0	2899	558	475	403	222	84	12	0	5	53	216	362	509		
	HDD18.3	5544	817	708	661	472	318	183	106	152	276	473	611	767		
	CDD10.0	408	0	0	0	0	24	81	161	114	27	1	0	0		
	CDD18.3	11	0	0	0	0	0	1	8	2	0	0	0	0		
	CDH23.3	223	0	0	0	0	12	47	129	34	1	0	0	0		
	CDH26.7	28	0	0	0	0	0	3	23	2	0	0	0	0		
(14) Wind	WSAvg	1.7	1.3	1.5	1.9	1.9	2.2	2.3	1.9	1.8	1.7	1.6	1.5	1.3		
(15) Precipitation	PrecAvg	608	39	26	26	34	45	67	94	84	51	55	45	41		
	PrecMax	875	69	54	48	84	88	122	233	151	113	162	98	76		
	PrecMin	461	5	1	10	7	12	18	17	27	14	15	16	6		
	PrecStd	99	16	13	9	19	20	25	51	36	25	33	21	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.7	7.1	9.8	18.5	25.0	27.1	29.3	26.3	20.9	14.3	10.4	6.8		
		MCWB	3.7	4.2	5.2	9.9	14.7	16.3	18.7	18.2	15.2	10.8	8.2	4.8		
	2%	DB	4.1	5.5	7.6	14.5	22.0	24.7	26.6	24.0	18.3	11.6	7.5	4.9		
		MCWB	2.5	3.2	4.0	8.3	13.4	15.2	17.4	17.1	13.9	9.7	6.2	3.2		
	5%	DB	2.4	3.8	5.9	11.9	19.1	21.8	24.5	21.9	16.6	10.3	5.6	3.5		
		MCWB	1.1	2.1	2.9	6.7	11.9	14.2	16.9	16.2	13.0	8.6	4.5	2.1		
	10%	DB	1.0	2.1	4.5	9.5	16.2	19.7	22.3	19.9	14.8	9.0	4.2	2.1		
		MCWB	0.1	0.6	2.0	5.2	10.7	13.3	15.9	15.3	12.0	7.3	3.3	1.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	4.5	5.7	10.8	16.0	18.0	20.0	19.7	16.2	11.6	8.5	5.1		
		MCDB	5.4	6.5	8.8	16.8	21.5	24.7	26.5	23.9	19.5	13.4	10.2	6.6		
	2%	WB	2.6	3.4	4.4	8.5	14.4	16.2	18.6	18.1	14.6	10.1	6.4	3.4		
		MCDB	3.9	5.2	7.2	14.0	20.6	21.6	24.4	22.1	17.4	11.2	7.4	4.7		
	5%	WB	1.3	2.1	3.3	7.1	12.7	15.0	17.6	16.8	13.4	8.9	4.7	2.2		
		MCDB	2.2	3.6	5.6	11.1	18.2	20.7	22.8	20.8	16.0	10.2	5.4	3.2		
	10%	WB	0.2	0.7	2.1	5.6	10.9	13.9	16.5	15.7	12.3	7.4	3.3	1.1		
		MCDB	0.9	1.9	4.2	9.0	15.7	18.8	21.2	19.1	14.4	8.8	4.1	1.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	9.5	10.2	11.2	10.7	10.6	10.3	9.0	6.6	5.2	6.7		
		MCDBR	7.6	7.9	9.7	15.4	17.2	15.4	15.0	13.9	11.6	8.5	6.5	7.2		
		MCWBR	6.9	6.5	6.7	9.6	9.4	7.6	7.4	7.7	6.6	5.7	6.3			
		MCDBR	7.3	7.9	9.4	14.1	15.4	13.3	13.0	11.9	10.4	8.0	6.4	6.9		
	5% WB	MCWBR	6.7	6.6	6.8	9.1	8.8	7.2	6.9	7.0	7.1	6.5	5.7	6.1		
(40) Clear-Sky Solar Irradiance	taub		0.220	0.281	0.307	0.343	0.347	0.343	0.357	0.348	0.317	0.301	N/A	0.348		
	taud		2.053	2.222	2.302	2.372	2.443	2.482	2.457	2.478	2.553	2.457	N/A	2.683		
	Ebn at Noon		416	659	794	840	869	880	853	830	795	649	N/A	246		
	Edh at Noon		30	60	82	96	97	96	96	86	66	49	N/A	18		
(44) All-Sky Solar Radiation	RadAvg		0.18	0.81	2.23	3.81	4.72	5.41	4.95	3.75	2.29	0.99	0.27	0.09		
	RadStd		0.02	0.12	0.24	0.31	0.50	0.38	0.54	0.32	0.26	0.16	0.05	0.01		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (42 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	-124	N/A	N/A	N/A

Nomenclature: See separate page